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Housing and Fertility

[PLACEHOLDER: motivation pitch — housing access shapes fertility]

- [PLACEHOLDER: motivation fact 1]
- [PLACEHOLDER: motivation fact 2]
- [PLACEHOLDER: motivation fact 3]

This Paper

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- [PLACEHOLDER: theory contribution]
- [PLACEHOLDER: quantitative contribution]
- [PLACEHOLDER: policy contribution]

Related Literature

- [PLACEHOLDER: Sommer–Sullivan–Verbrugge slot]
- [PLACEHOLDER: Doepke–Kindermann slot]
- [PLACEHOLDER: Greaney–Parkhomenko–Van Nieuwerburgh slot]
- [PLACEHOLDER: Couillard slot; Hacamo slot]

Environment

A two-period overlap: young households make family and housing choices, and become the old incumbents who hold the stock. A potential young household enters with income and wealth,

$$i = (y_i, a_i),$$

so who can reach owner housing is set at entry.

- Young households choose entry, tenure, how much space to occupy, fertility, and saving.
- Young owners carry their house into old age; young renters reach old age without one.
- Old incumbents choose consumption, how much housing to keep, and what to leave.

Selling triggers the capital-gains tax; dying forgives it through basis step-up. The wedge is

$$\bar{\ell} = \tau^g \frac{P \varpi}{1 + \varpi}.$$

The Family-Housing Cap

Rental and finance frictions collapse into one cap on family space. The owner cap is the tighter of the owner size limit and down-payment capacity:

$$h_i \leq H_i^O(q, \tau^P) = \min \left\{ \bar{h}^O, \frac{a_i \rho}{(1 - \phi)q} \right\}.$$

A renter faces $H_i^R = \bar{h}^R$, so tenure just maps different primitives into the same cap object. Substituting the cap into the child-cost condition prices the marginal child:

$$\underbrace{\frac{\vartheta(c_i^m - \chi n_i^m)}{n_i^m}}_{\text{willingness to pay for a marginal child}} = \underbrace{\chi + \kappa(q^m + \zeta_i^m)}_{\text{full price of a child}}.$$

A binding cap acts like a per-child tax of $\kappa \zeta_i^m$.

Access Gaps

The access gap is the goods value of one more unit of constrained family space. For a renter, the bundle reveals the bite of the cap:

$$\frac{u_{h,i}^Y}{u_{c,i}^Y} = \frac{\alpha(c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = q + \zeta_i^R.$$

A renter at the size cap values another unit of space above the rent it commands. For an owner, the gap separates finance from the size limit:

$$\zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i^O}(1 - \phi)P}_{\zeta_i^{O,F}, \text{ down-payment margin}} + \underbrace{\frac{\theta_i^O}{\lambda_i^O}}_{\text{size-limit margin}} \geq 0.$$

Either way ζ_i is computable from the observed consumption and housing bundle, net of child needs.

Where the Cap Binds, Fertility Responds

Under lifetime equal scaling, the fertility response to relaxing the cap is proportional to the same gap that measures access:

$$\psi_i^m = \frac{\kappa}{\alpha} \frac{\zeta_i^m}{c_i^m - \chi n_i^m} \Omega_i^m, \quad \Omega_i^m = \frac{\frac{\alpha}{h_i^m - \kappa n_i^m} + \frac{q^m}{c_i^m - \chi n_i^m}}{\frac{\vartheta}{(n_i^m)^2} + \frac{\chi^2}{(1+k)(c_i^m - \chi n_i^m)^2} + \frac{\alpha \kappa^2}{(h_i^m - \kappa n_i^m)^2}} > 0.$$

Holding exposure and pass-through fixed, a larger normalized gap means a larger birth response. So the households closest to the cap are the ones whose fertility moves.

Why the Stock Is Misallocated

Consider moving one unit from an old incumbent to a financially constrained young owner:

$$\underbrace{(q^O + \zeta_i^{O,F})}_{\text{buyer's marginal value}} - \underbrace{(q - \bar{\ell})}_{\text{incumbent's marginal value}} \\ = \zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell} > 0.$$

The market price cancels; what is left is the young buyer's financing gap plus the two tax wedges, and it is always positive. The problem is not that the old occupy housing, but that their lock-in sits next to young households with unmet access gaps.

Taking the Mechanism to a Lifecycle Model

- Lifecycle: 17 four-year periods, ages 18 to 82.
- Income and children: a 5-state income process; parity 0, 1, 2 with stochastic child stages.
- Tenure: renting is capped at 6 rooms; the owner ladder is $\{2, 4, 6, 8, 10\}$ rooms.
- Financing and market: the down payment is $(1 - \phi)$ with $\phi = 0.80$; a single housing market clears q in general equilibrium.

The tie back to the mechanism is the down-payment term: it enters the owner access gap $\zeta_i^{O,F}$ through $(1 - \phi)P$.

Calibration

- The fit is a 14-moment SMM table against 13 estimated parameters.
- Solve on a fine wealth grid, with a finer grid for final verification.¹
- Empirical inputs: room moments from the ACS, and lifecycle wealth and tenure moments from the PSID.
- The rental cap is not estimated. We set $\bar{h}^R = 6$ rooms from an external rule: the 90th percentile of rental unit sizes, following Greaney, Parkhomenko, and Van Nieuwerburgh (2025), measured in our own ACS sample.

The precise statement is that the p90 of renter sizes falls in the 6-room bin, so the maintained cap is a measured feature of the rental stock, not a free parameter.

¹Search at $N_b = 120$ nodes and verify at $N_b = 240$; a coarser grid tunes to numerical noise, so the reported fit is checked at the finer resolution.

The Data Do Not Fight the Cap

Re-solving at nearby cap values traces out a shallow valley around the external rule:

$\bar{h}^R$	best loss	status
5.5	7.69	converged
6.0	6.31	converged
6.5	6.24	still improving
7.0	7.15	preliminary

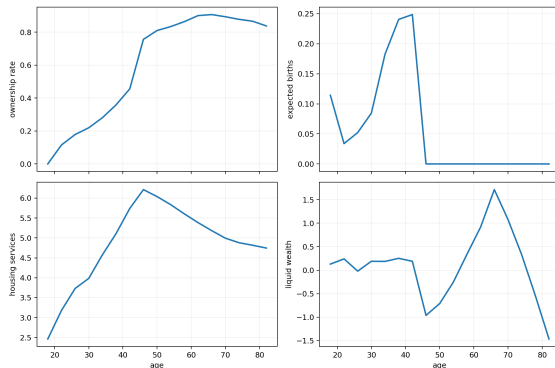
The data mildly prefer 6.5, but both flanks rise sharply, so the cap is identified and its minimum sits within half a room of the external value. We keep $\bar{h}^R = 6$ as the baseline and carry 6.5 as a robustness check.

Model Fit

moment	target	model	gap	contrib
tfr	1.918	1.947	+0.029	0.02
childless_rate	0.188	0.260	+0.072	0.10
own_rate	0.576	0.531	-0.045	0.20
own_family_gap	0.168	0.298	+0.131	0.77
housing_increment_0to1	0.664	0.617	-0.048	0.03
old parent-childless wealth gap 65–75	1.007	0.835	-0.172	0.06
childless renter mean rooms	3.805	3.656	-0.150	0.13
childless owner share rooms ≥ 6	0.596	0.595	-0.001	0.00
old nonhousing wealth/income median	2.231	2.328	+0.097	0.01
young renter wealth/income	0.179	0.354	+0.175	0.37
owner-renter mean rooms	2.419	2.152	-0.266	0.85
old_age_own_rate	0.764	0.892	+0.128	2.63
own_rate_2534	0.341	0.226	-0.115	1.06
parent 3+ vs 1–2 rooms gap	0.368	0.264	-0.104	0.09

The family-ownership moments fit well. Three misses carry the loss: old-age ownership is too high, young ownership and wealth are mistimed, and childlessness overshoots.

Lifecycle Profiles: A Deadline Purchase



- Ownership jumps at the last fertile age, not gradually before it.
- Births cluster at age 33 with a spike at the deadline; in the data they peak near 27.
- Liquid wealth drops sharply at the family purchase, as households lever up to the limit.

The remaining fit problem is timing: the model buys and gives birth too late.

What the Misses Are Telling Us

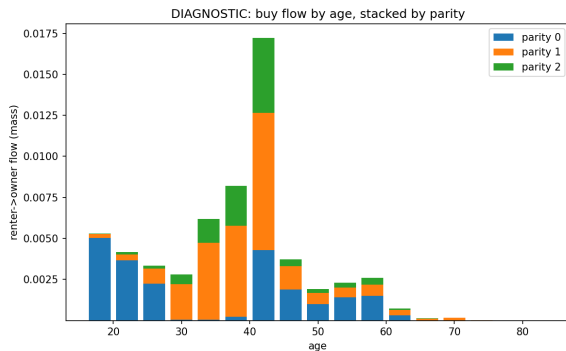
diagnosed miss	fix designed, not yet adopted
Old-age ownership is 0.892 against 0.764 in the data.	Give old owners a health, mortality, or downsizing exit.
Young ownership is 0.226 against 0.341, and young wealth is 0.354 against 0.179.	Fix fertility timing, rather than re-searching the same parameter box.
Childlessness by income runs the wrong way.	Add a child time cost, so children cost time as well as goods and rooms.
The misses hang together: late fertility and no old-age exit are doing work that no parameter can do cleanly.	

The Access Gap Shows Up in the Calibration

- Among fertile renters constrained by the rental size limit, $\zeta/q = 1.28$.
- These renters value family space at $2.3\times$ market user cost.
- A share 0.734 of the fertility-response mass sits on households with $\zeta > 0$.
- The share is robust to numerical resolution, to how the targets are weighted, and to the owner ladder.

This is the equal-scaling prediction in the calibrated model: fertility exposure rises with the normalized gap $\zeta_i^m / (c_i^m - \chi n_i^m)$.

Births Trigger the Purchase at the Deadline



- About 29% of the age-42 cohort buys in that single period.
- Seven in ten of these buyers are parents, and half have a newborn.
- Buyers arrive with slack: the average one is about 10 wealth-grid cells above its down-payment threshold.

The purchase margin is a birth-timing margin, not slow saving finally crossing the constraint.

Who Holds the Family-Sized Stock

The calibrated allocation gives the empirical face of the misallocation result:

holder group	share of rooms ≥ 6 stock
post-fertile households	87%
households with no kids at home	72%
fertile parents	12.6%

The households with the largest measured access gap hold almost none of the family-sized owner stock. This is the planner-equilibrium comparison, stated as an accounting fact about the calibrated economy.

A Price Instrument Capitalizes but Barely Moves Births

- Consider a 2% tax on large homes.
- Large-home prices fall by 12.3%.
- Fertility hardly moves: $\Delta \text{TFR} = +0.004$.
- The response is robust to numerical resolution, to the housing menu, and to the weighting of targets.

A pure price instrument capitalizes into house prices; it does not create births.

Birth-Linked Purchase Grants Move Fertility

A grant paid at a birth-linked purchase raises fertility, and the response grows with the size of the grant:

grant size	ΔTFR
0.1	+0.01
0.4	+0.13
0.58	+0.26
1.0	+0.49

Prices barely move — large-home prices rise about 1%.

- One caveat: a small conditional grant can pull purchases back before the birth, delaying them through anticipation.

Grants move the birth margin because they relax the owner financing gap $\zeta_i^{O,F}$.

A Package That Decentralizes the Reallocation

Tax the incumbent stock and spend the revenue on young buyers' access:

$$\text{large-home tax} + \text{birth-linked purchase grant} \Rightarrow \Delta \text{TFR} = +0.15, \quad \Delta p_{\text{large}} = -11\%.$$

The tax lowers what incumbents will pay to keep large units; the grant closes the young side's financing gap at the birth margin. Revenue-funded, it is the decentralized version of the planner's move — and it works through access, not price.

The Bridge, in Three Numbers

1. Access: fertile renters at the cap have $\zeta/q = 1.28$, and 0.734 of the fertility-response mass sits on households with $\zeta > 0$.
2. Allocation: 87% of the family-sized stock is held by post-fertile households.
3. Policy: the tax-funded access transfer raises TFR by 0.15 while cutting large-home prices 11%.

The same object ζ measures fertility exposure and stock misallocation, and the package is the decentralized version of the planner's reallocation.

Where This Version Falls Short

- Several parameters sit at their search bounds.²
- Fertility is mistimed: births cluster at 33, against a data peak near 27.
- Planned extension: a child time cost, so children are priced partly in foregone income.
- Planned extensions: an old-age exit margin, and childlessness targets consistent with the estimand.

These belong in the draft as honest limitations, not as settled calibration details.

²Eight of the 13 estimated parameters are at or against a bound at the current calibration — evidence that the parameter space is compensating for missing mechanisms, not a failure of the search.

Conclusion

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